

Innovating for National Development: Technological Innovation the vehicle to the future

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What is Innovation?

- In economic terms, innovation describes the development and application of ideas and technologies that improve goods and services or make their production more efficient.
- A classic example of innovation is the development of steam engine technology in the 18th century. Steam engines could be put to use in factories, enabling mass production, and they revolutionized transport with the railways. More recently, information technology transformed the way companies produce and sell their goods and services, while opening up new markets and new business models.

- *The European Union*

Why do we need Innovation?

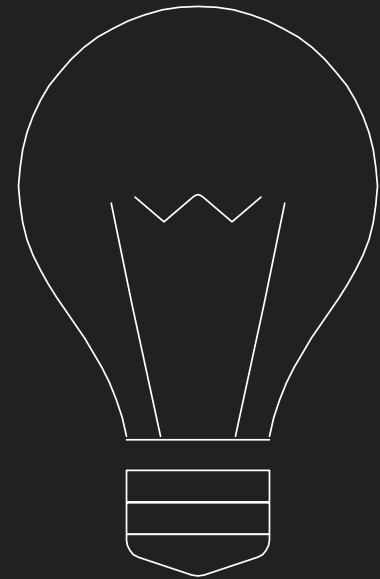
- One of the major benefits of innovation is its contribution to economic growth. Simply put, innovation can lead to higher productivity, meaning that the same input generates a greater output. As productivity rises, more goods and services are produced – in other words, the economy grows.

How can Innovation boost Development?

- Innovation can make a difference in addressing urgent developmental challenges such as providing access to drinking water, eradicating neglected diseases or reducing hunger. The transfer and, when necessary, adaptation of technologies developed in developed countries can often contribute significantly to these goals.

How Innovation leads to growth

- New ideas and technologies are developed and applied, generating greater output with the same input. More goods and services are produced, stimulating wages and business profitability.



How does innovation take full effect?

- Innovation usually starts on a small scale, e.g. when a new technology is first applied in the company where it has been developed. However, for the full benefits of innovation to be realised, it is necessary for it to spread across the economy and equally benefit companies in different sectors and of different sizes. Experts call this process the diffusion of innovation.

How can Innovation be promoted?

- Structural measures to promote innovation include increasing spending on research and development (R&D) and investing in education, as well as enabling entrepreneurs to start businesses more easily and for failed businesses to exit the market more quickly. In addition, companies can facilitate innovation by investing in their staff and conducting their own R&D.

Why innovation is important for developing and emerging countries

Country category	Mechanism/objective of innovation	Type/source of innovation and main agents involved
Developing/low- income countries and emerging and middle- income countries	- Adoption requires adaptation: Innovation needs to respond to specific "local" conditions for outcomes - Examples: New plant varieties for agri- culture, efforts at developing new methods for mineral extraction in the Chilean copper industry to satisfy local needs	- Incremental innovation based on foreign innovations and technologies - Agents: universities and research institutes, leading private businesses, esp. those with exposure to foreign markets and businesses
	- Inclusive innovation: for/by low- and middle-income households to improve welfare and access to business opportunities - Examples: India (nano cars; grassroots innovation), mobile banking services.	- Incremental innovation based on foreign technology and/or local, traditional knowledge generated "out of necessity" - Social innovation helping to introduce technical innovations in communities - Agents: NGOs, small firms, public and private associations engaged in disseminating knowledge via networks, private, often large businesses
Mainly middle- income countries but also some opportunities for developing /low- income countries	- Build up innovation capacities that will be key for reaching the world technological frontier in many industries, esp. relevant to avoid "middle-income traps" - Example: Korea increased R&D efforts in the 1990s	- Incremental and radical innovation capacity to compete with leading world innovators - Agents: Requires full development of innovation systems involving diasporas as a connector
	- Address environmental, health and social challenges through global innovation efforts and local efforts to address them - Example: Innovations concerning soil	- Major innovations and scientific research conducted in global partnerships but also marginal innovations to address welfare of poor people - Agents: Public and private universities and research institutions connected to global networks but also major private businesses operating in these sectors
	- Build-up niche competencies i.e. growth/exports in sectors of comparative advantage - Example: Colombian and Ecuadorian flower industry - Malaysia's palm oil sector	- Incremental innovations based on applying foreign innovations and technologies strategically to support industrial development - Agents: Public institutions to address co- ordination challenges, private sector initiative including foreign companies .../...

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Mainly emerging/ middle-income countries after initial progress on dimensions above	• Climb the value ladder in global value chains • Example: Automotive industry in Malaysia, India's software industry	• Incremental and radical innovation capacity to differentiate contributions • Agents: Involves private sectors with support from public agents, intermediaries, diasporas can play a central role, large firms can be important
	• Keep competitiveness in frontier industries when the country is already at the frontier • Example: Brazilian company Embraer as well as leading R&D firms from emerging economies	• Innovation is identical to developed countries exposed to developments in the global market • Agents: Involves mainly private sector in interaction with public research institutions and universities, global partnerships often equally of relevance, role of large firms
	• Build-up niche competencies i.e. growth/exports in sectors of comparative advantage • Example: Colombian and Ecuadorian flower industry • Malaysia's palm oil sector	• Incremental innovations based on applying foreign innovations and technologies strategically to support industrial development • Agents: Public institutions to address co-ordination challenges, private sector initiative including foreign companies .../...

BROOKINGS

SERIES: Foresight Africa



AFRICA IN FOCUS

Investment in science and technology is key to an African economic boom

Ameenah Gurib-Fakim and Landry Signé · Wednesday, January 26, 2022

Innovation for Development in Africa

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- The African continent represents 20 percent of the earth's surface and is home to 1.3 billion people—likely reaching 2.53 billion people by 2050. It boasts 60 percent of the world's arable lands, large swathes of forests, 30 percent of the world's reserve of minerals, and the youngest population of any continent. Yet, despite these riches, it produces only 3 percent of global GDP, accounts for less than 3 percent of international trade (mainly primary commodities and natural resources), and shoulders 25 percent of the global disease burden. The picture is particularly bleak when it comes to research and innovation: Africa contributes just 2 percent of world research output, accounts for only 1.3 percent of research spending, and produces 0.1 percent of all patents.

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Innovation for Development in Africa

- A lack of investment in science and technology has undermined Africa's economic transformation at both the structural level (the shift of workers and resources from low- to higher-productivity sectors) and the sectoral level (the growth of productivity within sectors). This lack of investment has had far-reaching consequences: Without the economic and scientific infrastructure necessary for innovation, the continent has continued to rely on the colonial development model of resource extraction, which is both unsustainable and largely responsible for its debilitating poverty and aid dependency. These challenges have been compounded by economic fragmentation, as smaller markets constrain the long-term investments and patient capital that would foster innovation and drive technology transfer in the context of globalization.

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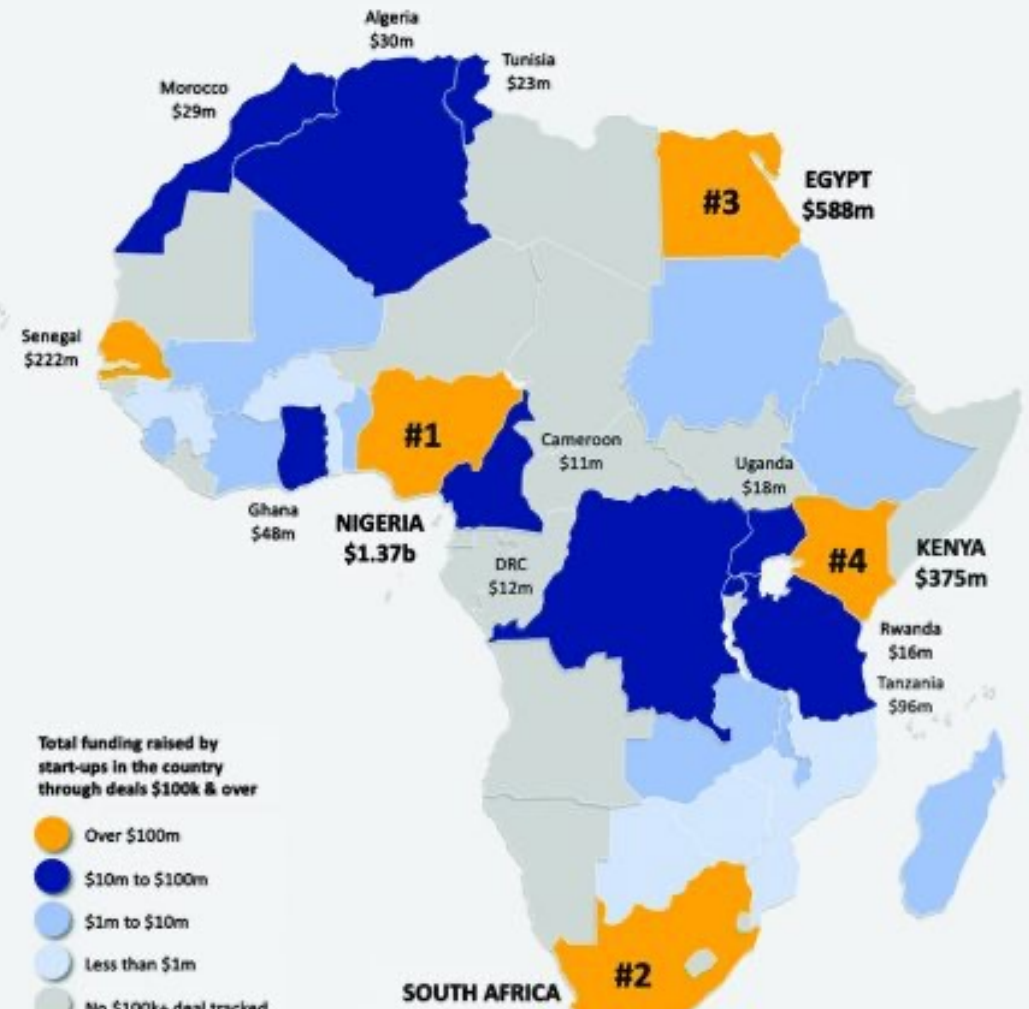
- The silver lining is that there is potential here with growing recognition by policymakers of the role that science and technology can play in achieving national development goals and transforming Africa's economic growth story. Moreover, given the positive correlation between growth and environments that beget competition and innovation, competitiveness must be fostered.
- The African tech ecosystem grew exponentially in 2021 with capital investments hitting all-time highs. Africa's young, digitally-literate population is fuelling this rapid technological development. The many successes of 2021 and host of fast-growing tech-companies across the continent is creating hope for more fantastic feats in years to come.

Innovation for Development in Africa

Total funding raised by start-ups in 2021

as of 29/11

For more, visit thebigdeal.substack.com by Max Cuvellier & Maxime Bayen



Innovation for Development in Africa

- Africa's tech ecosystem has experienced tremendous growth in recent years, with 2021 seeing a record number of investments.
- Tech start-ups across the continent received \$4,77 billion of investments, with the emergence of at least 5 new tech-unicorns (start-ups valued at \$1 billion).
- A key feature of this growth is tech talent development and the empowerment of female start-up leaders and developers.